Qidong Sun

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EDUCATION

Chongqing University — Elite Institute of Engineering *Sept. 2023 – Jun. 2027 (Expected)*
B.Eng., Robotics Engineering | GPA 3.51/4.00 | Weighted Average 86.00/100 | IELTS Academic 6.5 (overall)
Relevant Coursework: Mathematics and Physics III (96/100); Foundations of Robotics (94/100); Principles of Engineering (93/100); Quantitative Engineering Design II (92/100).

VISITING EXPERIENCE

Singapore University of Technology and Design (SUTD), Singapore *Sept. 2026 – Dec. 2026*
China Scholarship Council (CSC) fully funded visiting scholar.

RESEARCH EXPERIENCE

VLA-Based Dual-Arm Manipulation Pipeline via pi-0 in NVIDIA Isaac Sim
CQU Embodied Intelligence Lab *Sept. 2025 – Mar. 2026*

- Developed a dual-arm simulation pipeline in NVIDIA Isaac Sim to collect multimodal demonstrations with RGB observations, language instructions, and proprioceptive joint states.
- Implemented a WebSocket client connecting a pi-0 VLA model with the simulator for closed-loop observation–action communication.
- Automated conversion of simulation episodes into LeRobot-style formats and Parquet datasets, with validation for action–observation consistency, joint alignment, and trajectory logging.

Remote Research Internship on VLA Long-Horizon Memory
SJTU, advised by Prof. Zhipeng Zhang *Apr. 2026 – Sept. 2026*

- Investigated persistent task-state memory for VLA systems across retrieval-based, discrete, frozen-VLM history-conditioning, and recurrent GRU approaches.
- Built controlled evaluations with current-only, fixed-history, zero-memory, and shuffled-history baselines; recurrent memory reduced validation Action L1 from 0.143 to 0.095, while zeroed and shuffled memory degraded to 0.248 and 0.197.
- Diagnosed shortcut learning through ablations: despite 0.850 LOSO task-state accuracy, transition recognition remained weak and memory latents strongly encoded elapsed time rather than robust multi-event state.

PATENTS

Drive-Steering Wheel *Granted Sept. 2026*
Chinese Utility Model Patent, No. ZL 2025 2 2280198.X, Publication No. CN 224702860 U. **Co-inventor**.

SELECTED PROJECTS

RoboMaster Decision Stack Refactor (ROS 2, C++)
GitHub *Jul. 2024 – May 2025*

- Refactored the autonomy stack into state estimation, decision-making, and action execution layers, improving modularity and scalability.
- Co-designed and debugged ROS 2 communication interfaces with the electrical-control team, aligning message formats, control commands, and execution feedback.
- Iteratively tested and maintained the decision stack in simulation and on the robot, diagnosing integration issues and improving communication robustness.
- Tested navigation modules in Gazebo and ensured stable communication for real-robot deployment.

Unitree Go2 Quadruped Locomotion and Parkour Learning
SUTD, advised by Asst. Prof. Yihan Du and Dr. Shen Chen *Jun. 2026 – Present*

- Reproduced and adapted quadruped parkour pipelines in NVIDIA Isaac Gym for Unitree Go2, covering PPO-based locomotion, privileged teacher policies, and teacher–student learning for perception-based control.
- Trained Go2 walking and parkour-field teacher policies and analyzed obstacle-traversal behaviors and failure cases across simulated terrains.
- Migrated an Extreme Parkour pipeline from Unitree A1 to Go2 and validated robot configurations and training stability, preparing subsequent visual-policy distillation and deployment experiments.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB. **Robotics and simulation:** PyTorch, ROS 2, NVIDIA Isaac Sim, Gazebo, MuJoCo, LeRobot, RViz2, Nav2, SLAM Toolbox. **Tools:** Linux/Ubuntu, Git, CMake, WebSocket.

HONORS & AWARDS

- CSC fully funded visiting scholarship, SUTD, 2026
- National Third Prize, RoboMaster Robotics Competition (National final)
- Regional Second Prize, RoboMaster Robotics Competition, Central China Regional
- Regional Third Prize, RoboMaster University League, Southwest Station
- Successful Participant, Mathematical Contest in Modeling (MCM), 2026
- Participant, 4th IOTSC Summer Camp for Outstanding University Students, University of Macau, 2026